

Polynomial-Size Mass Production of Boolean Circuits

An exploratory transcript

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Abstract

For a Boolean function f , let $f^{\times t}$ evaluate f on t independent input blocks. This transcript asks for upper bounds on $C(f^{\times t})$ that use the actual, polynomial circuit size $C(f)$ and beat the replication bound $tC(f)$. There are three conclusions.

First, a theorem promising a strict saving for *every* polynomial-size function is false: functions of near-linear size are already forced to pay linear input cost on every copy. If $v(f)$ is the number of essential input variables, the right optimistic target is instead

$$C(f^{\times t}) = O(C(f) + tv(f)).$$

This target is optimal up to a constant against elementary lower bounds, but no unconditional proof is supplied here and none was found in the sources audited below.

Second, Lipton proves the slightly weaker $O(C(f) \log C(f) + tn)$ bound under the strong hypothesis that every polynomial-time multiple-output Boolean map has linear-size circuits. Applying his evaluator argument to the essential core replaces n by $v(f)$.

Third, there is an unconditional polynomial-size example with a genuine asymptotic saving. For infinitely many m , a scalar bilinear function on $2m$ essential bits has one-copy complexity $\Theta(m^2/\log m)$, whereas m independent copies have complexity $O(m^{\log_2 7})$. Thus the batch cost is $o(mC(f))$. The hard bilinear forms are obtained by counting, so this is an existence result rather than an explicit family or a theorem for all small circuits.

1 Verdict and model

1.1 What has actually been obtained

The exploration produced one unconditional positive theorem and a hierarchy of universal conjectures. The positive theorem is deliberately stated first so that “polynomial-size mass production” does not refer only to a wish.

Theorem 1.1 (Scalar polynomial-size mass production). *For every sufficiently large power of two m , there is an invertible matrix $A_m \in \mathbb{F}_2^{m \times m}$ for which the scalar Boolean function*

$$b_{A_m}(x, y) = y^\top A_m x, \quad x, y \in \mathbb{F}_2^m,$$

has all $2m$ variables essential and satisfies

$$C(b_{A_m}) = \Theta\left(\frac{m^2}{\log m}\right), \quad C(b_{A_m}^{\times m}) = O(m^{\log_2 7}).$$

Consequently,

$$\frac{C(b_{A_m}^{\times m})}{mC(b_{A_m})} = O\left(\frac{\log m}{m^{3-\log_2 7}}\right) = o(1).$$

The proof is in Section 5. This is a single-output Boolean function, not a vector-valued evasion of the question. Its input length is $2m$, its circuit size is quadratic up to a logarithm, and the number of copies is linear, so all parameters are polynomial. On the other hand, the matrix A_m is chosen nonconstructively. This is a scalarization of the standard matrix-multiplication mass-production example; no novelty or priority claim is intended.

Status: Proved. This note is standalone for the polynomial-size analysis. Its two references to `main.tex` mean the affine-line construction and the exponential-range mass-production theorem developed in that companion manuscript.

1.2 Circuit convention

Circuits use a fixed finite complete Boolean basis of fan-in at most two, such as $\{\wedge, \vee, \neg\}$. Size counts non-input gates. Constants, fan-out, wires, and output designations are free. XOR therefore has constant cost. Changing to any other fixed finite complete basis changes every bound below by at most a constant factor. The notation $\tilde{O}$ suppresses factors polylogarithmic in the parameters displayed in the same statement.

For $f : \{0, 1\}^n \rightarrow \{0, 1\}$ and $t \geq 1$, define the t -fold independent product

$$f^{\times t}(x^{(1)}, \dots, x^{(t)}) = (f(x^{(1)}), \dots, f(x^{(t)})).$$

The trivial upper bound is

$$C(f^{\times t}) \leq tC(f). \tag{1}$$

The question is when the ratio between the two sides tends to zero.

Status vocabulary. **Proved** means that a proof is included or a cited theorem was checked. **Conditional** means that the displayed conclusion depends on a stated unproved hypothesis. **Candidate** marks a proposed mechanism with a named proof obligation. **Open** means that no proof is claimed here.

2 The intrinsic scale of the problem

2.1 Essential arity

Definition 2.1 (Essential arity). An input coordinate of $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is *essential* if changing that coordinate alone can change f . Write $v(f)$ for the number of essential coordinates. Restricting the inessential coordinates and keeping one copy of every essential coordinate gives an *essential core* $f^{\text{core}} : \{0, 1\}^{v(f)} \rightarrow \{0, 1\}$.

Lemma 2.2 (Dummy-padding invariance). *For every f and $t \geq 1$,*

$$C(f) = C(f^{\text{core}}), \quad C(f^{\times t}) = C((f^{\text{core}})^{\times t}).$$

Proof. A circuit for the core computes f after its input wires are relabelled and the dummy pins are ignored. Conversely, fixing all dummy inputs in a circuit for f produces a circuit of no larger size for the core. Apply the same two operations independently to every input block. $\square$

Thus the ambient arity n can be inflated without changing the task. Every size-sensitive statement below uses $v(f)$.

Definition 2.3 (Individual mass-production curve).

$$\text{MP}_f(t) = C(f^{\times t}).$$

The elementary identities

$$\text{MP}_f(1) = C(f), \quad \text{MP}_f(t+u) \leq \text{MP}_f(t) + \text{MP}_f(u), \quad \text{MP}_f(t) \leq \text{MP}_f(t+1)$$

follow respectively from the definition, parallel composition, and fixing one input block and discarding its output. By subadditivity, Fekete's lemma also gives the well-defined asymptotic cost per copy

$$\overline{C}(f) := \lim_{t \rightarrow \infty} \frac{C(f^{\times t})}{t} = \inf_{t \geq 1} \frac{C(f^{\times t})}{t}.$$

2.2 A sharp elementary lower envelope

The usual input-incidence argument loses a small amount when outputs are allowed to be input wires. The following graph form handles that case.

Lemma 2.4 (Input-output incidence). *Let $F : \{0,1\}^N \rightarrow \{0,1\}^q$ be a vector-valued Boolean function, and suppose that V input coordinates are essential for at least one output. Then*

$$C(F) \geq V - q.$$

Proof. Append one cost-free sink to every designated output of a circuit for F . For each essential input, retain one directed path from that input to an output it influences, and let H be the union of these paths. If g_H is the number of gates retained in H , $r \leq q$ the number of retained output sinks, e the number of edges, and c the number of weak components of H , then every component contains a sink, so $c \leq r$. The underlying undirected graph gives

$$e \geq V + g_H + r - c.$$

At most $2g_H$ edges enter retained gates and exactly r enter sinks, so $e \leq 2g_H + r$. Hence $g_H \geq V - c \geq V - r \geq V - q$. The original circuit has at least g_H gates. $\square$

Proposition 2.5 (The unavoidable scale). *For every scalar Boolean function f , with $s = C(f)$ and $v = v(f)$, and every $t \geq 1$,*

$$\boxed{\max\{s, t(v-1)\} \leq C(f^{\times t}) \leq ts.} \quad (2)$$

If $v \geq 2$, then in particular

$$C(f^{\times t}) \geq \frac{s + tv}{3}.$$

Proof. Fixing $t-1$ blocks and retaining one output yields a circuit for f , giving the lower bound s . The vector function $f^{\times t}$ has tv essential inputs and t outputs, so Lemma 2.4 gives $t(v-1)$. The upper bound is replication. Finally, $v-1 \geq v/2$ for $v \geq 2$, whence $s + tv \leq 3 \max\{s, t(v-1)\}$. $\square$

At $t = 1$, the proposition also gives

$$v(f) \leq C(f) + 1. \quad (3)$$

2.3 Why a uniform strict saving is impossible

Proposition 2.6 (Exact and constant-factor no-go examples). *For $f(x_1, x_2) = x_1 \wedge x_2$ over a basis containing AND,*

$$C(f) = 1, \quad C(f^{\times t}) = t.$$

For parity on v inputs over $\{\wedge, \vee, \neg\}$,

$$t(v-1) \leq C(\text{PAR}_v^{\times t}) \leq 4t(v-1).$$

More generally, if every variable of f_v is essential and $C(f_v) = O(v)$, then $C(f_v^{\times t}) = \Omega(tC(f_v))$ uniformly in t .

Proof. For AND, the upper bound uses one gate per copy and the lower bound is $t(v-1) = t$. An XOR of two bits has a constant-size implementation, so a chain computes parity in at most $4(v-1)$ gates; the lower bound again follows from Proposition 2.5. If $C(f_v) \leq Kv$, then $t(v-1) \geq tv/2 \geq tC(f_v)/(2K)$ for $v \geq 2$. $\square$

Therefore a claim of $o(tC(f))$ for *all* polynomial-size functions is literally false. Along any asymptotic family with $v(f) \rightarrow \infty$ and $C(f) > 0$, a strict saving can occur only if

$$t \rightarrow \infty \quad \text{and} \quad \frac{C(f)}{v(f)-1} \rightarrow \infty, \quad (4)$$

equivalently $C(f)/v(f) \rightarrow \infty$ in this regime.

2.4 The right two targets

Conjecture 2.7 (Pointwise optimal mass production). *There is a basis-dependent constant K such that, for every scalar Boolean function f and every $t \geq 1$,*

$$C(f^{\times t}) \leq K(C(f) + tv(f)). \quad (5)$$

Status: Open. Proposition 2.5 shows that the right side is optimal up to a factor of three. It would give

$$\frac{C(f^{\times t})}{tC(f)} = O\left(\frac{1}{t} + \frac{v(f)}{C(f)}\right),$$

so the necessary conditions in (4) would also be sufficient.

The first milestone allows the natural cost of describing a circuit.

Conjecture 2.8 (Description-loss mass production). *For every scalar Boolean function f and every $t \geq 1$,*

$$C(f^{\times t}) = O(\min\{tC(f), C(f) \log(C(f) + 2) + tv(f)\}). \quad (6)$$

Status: Open unconditionally. Writing $s = C(f)$, this is a strict saving when $t/\log(s+2) \rightarrow \infty$ and $s/v(f) \rightarrow \infty$. If $s = \Theta(v^a)$ and $t = \Theta(v^b)$ with $a \geq 1$ and $b \geq 0$, the optimal target has size

$$O(v^a + v^{b+1})$$

and improves on v^{a+b} by a factor $\Omega(v^{\min\{b, a-1\}})$. The description-loss target gives the same polynomial phase diagram, with a logarithmic loss. Thus both predict a strict saving exactly when $a > 1$ and $b > 0$.

3 Three different meanings of a universal theorem

Fix a topologically ordered gate-list encoding. A circuit with v inputs and s gates has a description of length

$$d(v, s) = O((s + 1) \log(v + s + 2)). \quad (7)$$

For a minimum scalar circuit with at least two essential inputs, (3) reduces this to $O(s \log(s + 2))$.

There are three logically distinct goals.

- (i) *Pointwise existence*: for every f , some small circuit computes $f^{\times t}$, with the bound expressed using the minimum $C(f)$.
- (ii) *Effective compilation*: an algorithm receives a particular circuit Q of size s and t in unary, and outputs a small circuit for $(f_Q)^{\times t}$. The bound is now in terms of the supplied s , not the unknown minimum $C(f_Q)$.
- (iii) *Universal batch evaluation*: one circuit $U_{v,s,t}$ receives the description of any size- s circuit together with t data blocks and returns all t evaluations.

An effective universal evaluator gives an effective compiler by hardwiring the program, and a compiler gives pointwise existence by applying it to a minimum circuit as a logical witness. A merely nonuniform universal circuit need not be constructible, and separate compiled circuits need not assemble into one small universal evaluator. No reverse implication is automatic.

The $s \log s$ cost belongs naturally to the live program input of the third formulation. It is not an information-theoretic lower bound on the circuit left after the program is hardwired.

3.1 Lipton's conditional theorem

Lipton's exact statement uses the declared arity n . Lemma 2.2 lets us apply the same argument to the essential core and use $v(f)$ instead. The following restricted form of Lipton's hypothesis is sufficient for that evaluator argument; it is extremely strong and unproved.

Hypothesis $\mathbf{H}_{\text{func}}$. Every polynomial-time uniform family of Boolean maps $G_N : \{0, 1\}^N \rightarrow \{0, 1\}^{m(N)}$, where $m(N) \leq N$, has circuits of size $O(N)$. The hidden constant may depend on the family G .

Theorem 3.1 (Derived from Lipton's evaluator argument [4]). *Assume $\mathbf{H}_{\text{func}}$. Then, for every scalar Boolean function f and every $t \geq 1$,*

$$C(f^{\times t}) = O(tv(f) + C(f) \log(C(f) + 2)).$$

The same argument yields a nonuniform universal batch evaluator of size $O(tv + d(v, s))$ for size- s circuits. It does not supply an algorithm that constructs those evaluator circuits.

Proof. First handle $v \leq 1$ by replication. For $v \geq 2$, choose a standard fixed-width gate-list field of exact padded length $D(v, s) = \Theta((s + 1) \log(v + s + 2))$. Define one canonical function $E_N : \{0, 1\}^N \rightarrow \{0, 1\}^N$. Its input contains a self-delimiting header (v, s, t, D) , a D -bit circuit field, t blocks of v data bits, and padding to total length

$$N = tv + D + O(\log(v + s + t + D + 2)) = \Theta(tv + D),$$

where the hidden constants come only from the fixed header format. On a valid description, E_N simulates the encoded circuit in topological order on every block, writes the t answers in its first t output positions, and fills the rest with zeroes. Invalid encodings map to zero. Here $s \leq D \leq N$ and $t \leq tv \leq N$, so naive $O(ts)$ simulation takes at most $O(N^2)$ time. Thus the E_N form one fixed polynomial-time vector-function family.

Hypothesis $\mathbf{H}_{\text{func}}$ gives E_N circuits of size $O(N)$. Hardwire the header and padding to obtain a nonuniform universal evaluator of size $O(tv + d(v, s))$. Finally hardwire a minimum circuit for f and use (7) and (3). Constants and one-variable functions were handled at the start. $\square$

Status: Conditional. The evaluator algorithm is explicit; the small circuit implementing it is conditional and nonconstructive. Valiant’s universal circuits [11] explain the ordinary $O(s \log s)$ programmable-simulation cost, but do not themselves remove the factor t from t evaluations.

4 What the worst-case mass-production theorems do not yet give

The checked classical results answer the question when the one-copy function is allowed to have worst-case exponential complexity. They do not give the desired dependence on an actual polynomial value $s = C(f)$. This is a claim about the sources audited below, not an exhaustive no-result claim.

4.1 Checked baseline bounds

Shannon’s counting lower bound and Lupanov’s synthesis upper bound give the worst-case one-output scale

$$\max_{f: \{0,1\}^v \rightarrow \{0,1\}} C(f) = \Theta(2^v/v) \quad (8)$$

for any fixed complete finite basis; see [7, 5, 12]. Against that scale, the following are genuine mass-production results.

- Let $L_t(v) = \max_{f: \{0,1\}^v \rightarrow \{0,1\}} C(f^{\times t})$. In a fixed complete weighted basis with Lupanov constant ρ , Uhlig proved that every sequence $t = t(v)$ satisfying $\log t = o(v/\log v)$ obeys

$$L_t(v) \sim \rho \frac{2^v}{v}.$$

Consequently the $O(2^v/v)$ upper bound holds uniformly for every f , but the matching lower bound is worst-case only and does not compare the batch size with an individual function’s $C(f)$. A modern restatement and proof overview appear in [2]; see also [10, 9, 12].

- Paul’s Lemma 2 [6] gives, in his arbitrary-binary-gate convention,

$$C(f^{\times t}) = O\left(\max\{v2^v, vt \log^2(t+2)\}\right). \quad (9)$$

A constant-factor basis simulation transfers the estimate to the present model.

- The companion construction in `main.tex` proves the order-of-growth bound

$$C(f^{\times t}) = O_\gamma(2^v/v) \quad (t \leq 2^{\gamma v})$$

for every fixed $\gamma < 1$. Unlike Uhlig’s smaller-range theorem, that argument does not retain the optimal one-copy leading constant.

Paul's bound also explains the known polynomial amortized cost when the copy count is allowed to become very large. Partition requests into blocks of $k_0 = \lceil 2^v/v^2 \rceil$ and apply (9) to every block. This gives the following derived consequence, valid for every f and t :

$$C(f^{\times t}) = O(v^3 t + v2^v). \quad (10)$$

Thus the asymptotic cost per copy is $O(v^3)$. The setup term is still exponential, so this only helps after an exponential number of requests.

Two other statements in Paul's paper must be kept separate. His scalar two-copy theorem gives a small circuit for the *single-output* function $f(x) \vee f(y)$; it is not a circuit for the two-output map $f \times f$. His genuine two-output direct-product theorem is for vector-valued functions. Paul explicitly records that the corresponding nontrivial scalar $f \times f$ saving was not obtained there.

Albrecht's short paper [1] studies simultaneous realization. Its publisher preview and abstract promise small size and depth overhead for the corresponding Shannon functions. This transcript does not use a sharper formula from that paper: a theorem-level primary-source comparison has not been recovered here. This omission is intentional rather than an invitation to infer a bound.

4.2 The polynomial-size gap

Combining only the checked generic estimates gives the envelope

$$C(f^{\times t}) \leq \min \left\{ tC(f), O\left(\max\{v2^v, vt \log^2(t+2)\}\right), O(v^3 t + v2^v) \right\}, \quad (11)$$

with the $O(2^v/v)$ plateau available in the copy ranges just described. If both $C(f)$ and t are polynomial in v , every displayed generic alternative to replication retains an exponential setup term. Hence these theorems alone return $tC(f)$ as the useful bound in precisely the regime of interest. They do not refute a size-sensitive theorem; they simply do not interpolate to the actual value of $C(f)$.

4.3 Why ordinary counting does not prove a direct sum

The number of fan-in-two circuits of size at most q with N inputs and one output is

$$2^{O((q+1) \log(N+q+2))}. \quad (12)$$

Indeed, for every topologically ordered gate one chooses a constant-size gate type and at most two predecessors among $N + q + 2$ available sources, followed by one output source. With r designated outputs the corresponding upper count is

$$2^{O((q+r+1) \log(N+q+2))}, \quad (13)$$

because one must also list the r output sources.

The map $f \mapsto f^{\times t}$ is injective, but it introduces no new choice:

$$\left| \{f^{\times t} : f : \{0, 1\}^v \rightarrow \{0, 1\}\} \right| = 2^{2^v},$$

the same cardinality as the one-copy family. Likewise, the products of functions having circuits of size below $2s$ form a set of size at most $2^{O((s+1) \log(v+s+2))}$, with no factor t in the exponent. Applying (13) to candidate batch circuits adds an output-list term $t \log(tv + q + 2)$ to the *circuit* count but adds no entropy to the target family; this only weakens a counting lower bound. In parameter ranges where output listing is not already dominant, plain Shannon counting can force an s -scale lower bound, but it cannot manufacture a ts -scale direct sum merely by replacing every target with its t -fold product. A stronger lower bound for a particular f would need structural information, not just target-set entropy.

5 An unconditional scalar pilot theorem

This section proves Theorem 1.1. The construction is the cleanest evidence found here that polynomial-size scalar functions can exhibit the desired phenomenon.

5.1 The standard vector warmup

For a fixed matrix $A \in \mathbb{F}_2^{m \times m}$, the linear map $x \mapsto Ax$ has complexity $\Omega(m^2/\log m)$ for almost every A by counting. Evaluating it on m vectors at once is the matrix product AX , where the requested vectors are the columns of X . Fast matrix multiplication computes this in $O(m^\omega)$ Boolean gates for some $\omega < 3$; this standard example is recalled in [2]. The only defect for the present question is that $x \mapsto Ax$ has m output bits. A bilinear outer shell turns it into a scalar function without destroying the saving.

5.2 One copy is genuinely quadratic

For $A \in \mathbb{F}_2^{m \times m}$, put

$$b_A(x, y) = y^\top Ax = \bigoplus_{i,j \in [m]} A_{ij} y_i x_j.$$

Lemma 5.1 (Almost every invertible form is hard). *There is a constant $c > 0$ such that, for all sufficiently large m , all but an exponentially small fraction of the matrices in $\text{GL}(m, 2)$ satisfy*

$$C(b_A) \geq c \frac{m^2}{\log m}.$$

Every such b_A has all $2m$ variables essential.

Proof. The map $A \mapsto b_A$ is injective, because $b_A(e_j, e_i) = A_{ij}$. Moreover,

$$|\text{GL}(m, 2)| = 2^{m^2} \prod_{j=1}^m (1 - 2^{-j}) \geq c_0 2^{m^2}$$

for an absolute constant $c_0 > 0$. By (12), the number of functions on $2m$ inputs computed with at most $q = cm^2/\log m$ gates is $2^{O(cm^2)}$. Choosing c sufficiently small makes this $2^{m^2 - \Omega(m^2)}$, hence a vanishing fraction of the invertible forms.

If A is invertible, every row and every column is nonzero. A nonzero entry A_{ij} witnesses that x_j is essential after $y = e_i$ and the other x -coordinates are fixed to zero; symmetrically it witnesses that y_i is essential after $x = e_j$. $\square$

Lemma 5.2 (Four-Russians upper bound). *For every $A \in \mathbb{F}_2^{m \times m}$,*

$$C(b_A) = O(m^2/\log m).$$

Proof. Let $k = \lfloor \log_2 m \rfloor$ and partition the coordinates of x into $O(m/k)$ blocks of length at most k . For one block, all 2^k linear forms in its bits can be generated using $O(2^k)$ XOR gates: recursively retain all forms on the first $k-1$ bits and XOR each of them with the last bit. Across all blocks this costs

$$O((m/k)2^k) = O(m^2/\log m).$$

For every row of A , select the required precomputed form from each block and XOR the $O(m/k)$ selected values. All m rows of Ax cost another $O(m^2/k)$. Finally compute the m products $y_i(Ax)_i$ and XOR them, at cost $O(m)$. Replacing each XOR with a constant number of basis gates proves the claim. $\square$

Together, Lemmas 5.1 and 5.2 show that almost every invertible A has $C(b_A) = \Theta(m^2/\log m)$.

5.3 Many copies become one matrix product

Proposition 5.3 (Batching bilinear forms). *For every fixed $A \in \mathbb{F}_2^{m \times m}$ and every $t \geq 1$,*

$$C(b_A^{\times t}) = O\left(\left\lceil \frac{t}{m} \right\rceil m^{\log_2 7} + tm\right).$$

Proof. First take $t = m$. Arrange the x -inputs as the columns of an $m \times m$ matrix X . Strassen's seven-product recursion [8] computes $Z = AX$ over $\mathbb{F}_2$ using $O(m^{\log_2 7})$ field operations. It remains to compute $y_r^\top Z_{*,r}$ for every column r , which takes $O(m)$ AND/XOR gates per column and $O(m^2)$ in total. Since $\log_2 7 > 2$, the matrix multiplication dominates. Hardwiring A into a circuit for general matrix multiplication cannot increase its size.

For arbitrary t , divide the requests into groups of at most m , pad the last group with zero columns, and apply the same circuit to every group. The inner products over all genuine requests cost $O(tm)$. $\square$

Choosing a hard invertible A_m from Lemma 5.1 and setting $t = m$ proves Theorem 1.1. More generally, (5.3) yields a strict saving for every matrix family for which

$$C(b_{A_m}) = \omega(m^{\log_2 7 - 1})$$

and every sequence $t = t(m) \geq m$. Indeed, $\lceil t/m \rceil \leq 2t/m$, so the ratio in question is

$$O\left(\frac{m^{\log_2 7 - 1} + m}{C(b_{A_m})}\right) = o(1).$$

Counting supplies many such matrices, but proving that bound for an explicit A_m would itself prove an unrestricted superlinear circuit lower bound. No such lower bound is proved here. The nonconstructivity is therefore a real boundary, not a cosmetic omission.

6 Mechanism audit for a theorem about every small circuit

The bilinear theorem identifies a successful mechanism: a large amount of one-copy work is a fixed linear transformation, and different instances can be mixed and later separated by a fast algebraic algorithm. An all-functions theorem would need either to find an analogue of that structure in every superlinear circuit or to use a different kind of global reorganization.

6.1 Gate-by-gate SIMD is not mass production

Given a size- s circuit Q , one can share its topology while maintaining a length- t vector of values at every gate. At the bit-gate level this still uses $\Theta(t)$ gates for each genuinely binary operation, hence $\Theta(ts)$ gates overall. Proposition 2.6 shows that even $\text{AND}_2^{\times t}$ costs exactly t . Sharing a wiring diagram is not the same as sharing the values computed on independent inputs.

6.2 Macro tabulation needs small interfaces

A tempting Four-Russians plan partitions Q into k -gate macros, tabulates each macro, and answers all requests by table lookup. Let b be the number of live input wires crossing into a macro and r the number of values leaving it. An arbitrary b -input, r -output table already has description scale $r2^b$, and the t requests contain tb independent live input bits. If $b = \Theta(k)$, tabulation is exponentially more expensive than the macro it replaces. Choosing $k \asymp \log t$ does not by itself fix the problem: the boundary cost and the per-query routing must both be charged.

There is a precise restricted criterion behind this idea. Partition the gates of Q into modules M_i compatible with the quotient DAG, and let $\phi_i : \{0, 1\}^{b_i} \rightarrow \{0, 1\}^{r_i}$ be the function taking the distinct live values entering M_i to the distinct values leaving it. Module substitution gives

$$C(f^{\times t}) \leq \sum_i C(\phi_i^{\times t}). \quad (14)$$

Indeed, replace every module, in quotient-topological order, by a circuit for t independent copies of its interface map, keeping instance indices aligned on every connecting wire. This observation is exact, but replication on the right side gives $C(\phi_i^{\times t}) \leq tC(\phi_i) \leq t|M_i|$ and hence recovers the upper bound ts . Decomposition only helps if a substantial collection of the interface maps is itself batchable.

Load-bearing lemma. A decomposition proof would need a statement that every *minimum* superlinear circuit has enough large pieces with boundary much smaller than their internal size, together with a batch-lookup circuit whose total routing cost is below tk per piece. Bounded-degree layered expanders, oriented and reduced to fan-in two by constant-size gadgets, are a natural adversarial test: any proposed lemma must explain why their medium-size pieces do not expose linear boundary. No general decomposition lemma is proved here.

6.3 Restriction coding is not automatically size-sensitive

The construction in `main.tex` reorganizes the truth table of an arbitrary function by coding many prefix restrictions and recovering requested ones along affine lines. Its resource functions are restrictions and linear combinations of restrictions on a shorter suffix. Worst-case synthesis charges them by their suffix arity and gives the desired $2^v/v$ scale.

Starting instead from a size- s circuit does not give a corresponding bound on the *sum* of the minimum sizes of all those derived resource functions. Synthesizing every restriction separately can cost $2^p s$ after fixing a p -bit prefix, before the scheduling recursion even begins. The missing statement is therefore not another choice of code parameters; it is a restriction-sharing theorem that relates the aggregate complexity of a decodable resource family to the original s .

Even the simplest path-difference charge fails exponentially.

Proposition 6.1 (Restriction differences need not charge to the circuit). *For every sufficiently large k , there are $m = 32k$ and a linear map $G : \mathbb{F}_2^k \rightarrow \mathbb{F}_2^m$ such that the polynomial-size scalar function*

$$F_G(a, z) = \langle Ga, z \rangle, \quad a \in \mathbb{F}_2^k, \quad z \in \mathbb{F}_2^m, \quad C(F_G) = O(km),$$

has the following property. Write $(F_G)_a(z) := F_G(a, z)$ for a prefix restriction. For every ordering π of its 2^k prefix restrictions,

$$\sum_{i=1}^{2^k-1} C((F_G)_{\pi(i-1)} \oplus (F_G)_{\pi(i)}) \geq (2^k - 1)(m/4 - 1).$$

Proof. A random $m \times k$ binary matrix has, with positive probability, $\text{wt}(Gu) \geq m/4$ for every nonzero u : for fixed u , Gu is uniform, a Chernoff bound gives failure probability at most $e^{-m/16}$, and a union bound over fewer than 2^k choices succeeds when $m = 32k$. Compute Ga and then its inner product with z to get the $O(km)$ upper bound.

The difference between restrictions at distinct a, a' is the parity with coefficient vector $G(a - a')$. It depends on at least $m/4$ variables, so (3) gives circuit size at least $m/4 - 1$. Every consecutive pair in an ordering is distinct, proving the sum. $\square$

This refutes a universal assertion that consecutive restriction differences have total complexity $\text{poly}(k)C(f)$. On the other hand, the k functions $h_j(z) = \langle Ge_j, z \rangle$ generate the restrictions by $(F_G)_a = \bigoplus_{j:a_j=1} h_j$ and have aggregate size $O(km)$. If the j th address bit is one in all t requests, however, h_j is demanded on all t unrelated suffixes. Thus a viable code must control resource complexity and request congestion simultaneously.

Status: Candidate. Here is a formal template for the missing lemma. For a split input $x = (a, z) \in \{0, 1\}^p \times \{0, 1\}^{v-p}$, an XOR restriction scheme consists of resource functions $\mathcal{H} = \{h_1, \dots, h_M\}$ and, for every address a , a nonempty collection $\mathcal{R}(a)$ of recovery sets such that

$$f(a, z) = \bigoplus_{j \in R} h_j(z) \quad \text{for every } R \in \mathcal{R}(a).$$

For t runtime addresses, a selector chooses $R_i \in \mathcal{R}(a_i)$; its load on resource j is $d_j = |\{i : j \in R_i\}|$. The selector must also route the corresponding suffixes into fixed-capacity resource circuits and route their answers back to the t decoders.

Question 6.2 (Candidate restriction-resource budget). *For some growing $1 \leq p = p(s, t) \leq v$, does every size- s function admit an XOR restriction scheme, load caps $D_1, \dots, D_M$, and a selector/router of size $\tilde{O}(tv)$ such that, for every tuple of t runtime addresses, it chooses recoveries with $d_j \leq D_j$ and*

$$\sum_{j=1}^M C(h_j^{\times D_j}) = \tilde{O}(s + tv)?$$

The quantifiers “every f ” and “aggregate” are essential. A proposal that bounds each $C(h)$ separately by s merely recreates the factor being removed. Multiplexer circuits with many unrelated branches and circuits whose restricted gate functions rapidly diversify are immediate stress tests for Question 6.2.

6.4 Circuit-as-static-data compilation

The most conceptually direct route is to treat the circuit description as a one-time database and the t input blocks as queries. This is exactly the universal evaluator in Section 3. Lipton’s argument proves that a linear-circuit hypothesis for polynomial-time maps would compress the whole computation to

$$O(s \log s + tv).$$

Unconditionally, ordinary universal circuits charge roughly $s \log s$ for *each separately simulated evaluation* unless some additional batch structure is found. The P-completeness of the circuit value problem [3] is a warning about uniform parallel evaluation, but it is not a nonuniform circuit lower bound and therefore is not evidence against Conjectures 2.7 or 2.8.

6.5 Algebraic cores and other restricted classes

Theorem 1.1 suggests looking for circuits with an expensive static algebraic core: linear maps, bilinear maps, or low-degree tensor contractions followed by only $O(v)$ instance-specific work. Fast rectangular matrix multiplication could extend the copy range in such classes. What is missing for general Boolean circuits is a normal-form theorem that exposes a sufficiently large algebraic core without increasing the circuit size by the savings one hopes to obtain.

Other plausible pilot restrictions are small separator number, bounded congestion decompositions, or an explicit promise that every large module has polylogarithmic interface. Results for those classes would be informative, but they should be stated in terms of the promised structural parameter; they would not by themselves establish a theorem for arbitrary size- s DAGs.

7 The next theorem to try

The clean all-parameter conjectures may be stronger than necessary for a first result. The following polynomial-band formulation isolates a clean polynomially separated regime in which a uniform asymptotic saving is possible.

Conjecture 7.1 (Polynomial-band milestone). *For every fixed $A > 1$, $B > 0$, and $0 < \varepsilon < \min\{A - 1, B\}$, there is a constant $K = K(A, B, \varepsilon)$ such that the following holds for all sufficiently large v . If every input of $f : \{0, 1\}^v \rightarrow \{0, 1\}$ is essential,*

$$v^{1+\varepsilon} \leq s = C(f) \leq v^A, \quad v^\varepsilon \leq t \leq v^B,$$

then

$$C(f^{\times t}) \leq K(s \log(s + 2) + tv).$$

This would imply uniformly over the promised band that

$$\frac{C(f^{\times t})}{ts} = O\left(\frac{\log v}{v^\varepsilon} + \frac{1}{v^\varepsilon}\right) = o(1).$$

It is weaker than Conjecture 2.8, avoids the regimes where the elementary lower bounds forbid a saving, and directly answers the polynomial-size question. It is still **Open**.

The most concrete next candidate is Question 6.2. It has three advantages over the raw conjecture: it interfaces directly with the already developed recovery-code construction; it exposes every cost that must be paid; and it has obvious counterexample searches. A useful first experiment would compute, for small optimized circuits and several prefix sets, the number and aggregate circuit size of distinct restricted gate functions. A negative experiment would not refute all mass production, but it would quickly kill the current route.

8 Results ledger

Status	Statement	Evidence or boundary
Proved	$\max\{C(f), t(v(f) - 1)\} \leq C(f^{\times t}) \leq tC(f)$	Proposition 2.5; the lower bound is a fan-in graph count.
Proved	No theorem can guarantee a strict saving for every polynomial-size function	AND ₂ is exact and every essentially v -ary, $O(v)$ -size family has a constant-factor direct sum; Proposition 2.6.
Proved	Scalar polynomial-size mass production exists	Almost every invertible bilinear form has size $\Theta(m^2 / \log m)$, and m copies cost $O(m^{\log_2 7})$; Theorem 1.1.
Proved	Worst-case truth-table mass production	Uhlig, Paul, and the companion theorem give strong exponential-size plateaux, but their checked bounds retain exponential setup in the polynomial-size regime.
Conditional	$O(C(f) \log C(f) + tv(f))$ for every f	Theorem 3.1, assuming $\mathbf{H}_{\text{func}}$. This proves nonuniform existence, not an effective compiler.
Open	$O(C(f) + tv(f))$ for every f	Conjecture 2.7; pointwise optimal up to a constant against the elementary lower envelope.
Open	An unconditional polynomial-band theorem	Conjecture 7.1; the proposed next route is the restriction-resource budget in Question 6.2.

Bottom line. This transcript proves a polynomial-size mass-production theorem for a large nonconstructive family of scalar Boolean functions, with a polynomial number of copies and a vanishing ratio to replication. The genuinely interesting universal theorem is left open in this transcript, and no proof was found in the audited sources. Its plausible optimal form is $O(C(f) + tv(f))$; its first description-sensitive form is $O(C(f) \log C(f) + tv(f))$. Any proof has to reorganize computations across independent inputs, not merely share circuit syntax.

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